

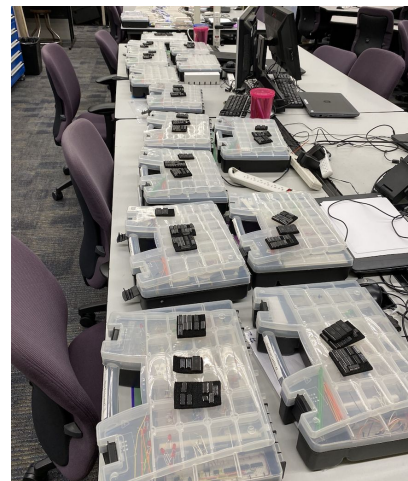
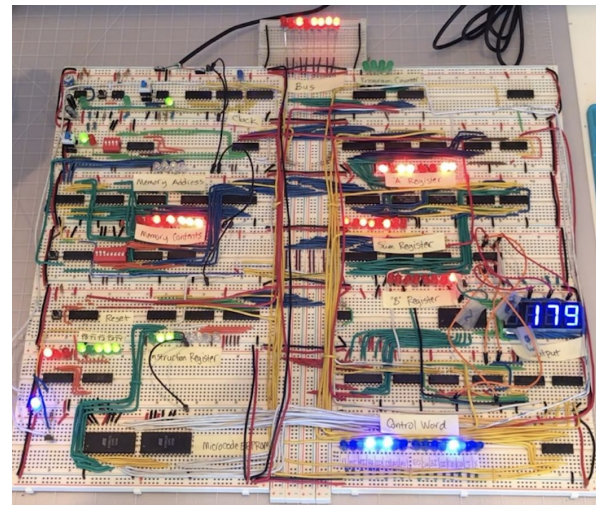
Regulatory Challenges and Advances:

Open EUAs, LAMP the future on an open testing ecosystem

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My CTE Classroom



Regulatory Pathways

- 1) FDA EUA – Emergency Use Authorization (IVD)
 - LDT for labs and IVD for manufacturers
- 2) CLIA Lab LDT – Lab Developed Test
- 3) IRB – Internal Review Board
 - typically at universities
 - needed for EUAs
- 4) Non-Dx Surveillance
 - not under CLIA
 - no "individual diagnostic" results
 - reflex positives to CLIA test

Regulatory Challenges & Hurdles

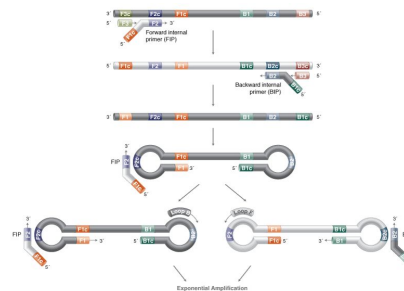
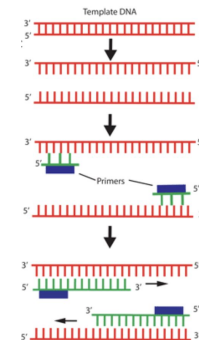
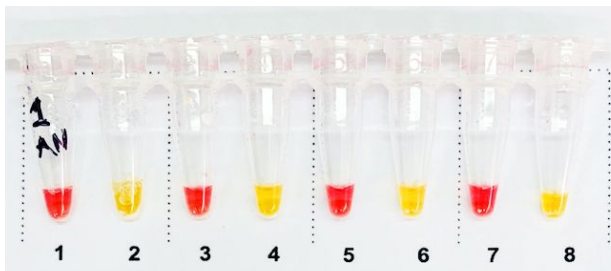
- Complexity of regulatory space
- FDA approves test systems not protocols
- Accessing clinical samples
- Performing clinical studies
- Submission paperwork
- Changing priorities

Open EUAs

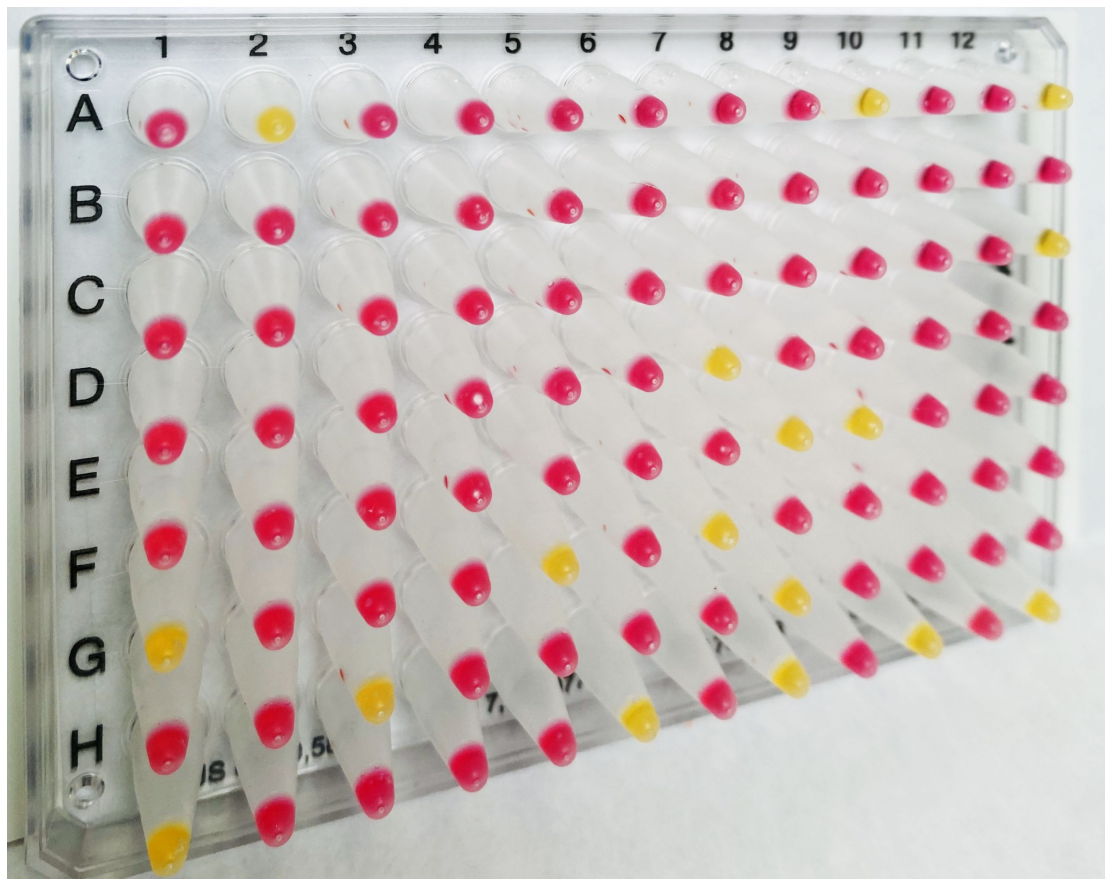
	 Typical IVD EUA	 CDC EUA	 Yale SalivaDirect™	 SHIELD	 FloodLAMP EasyPCR™	 FloodLAMP QuickColor™
Disclosure of all chemicals and reagents?	No	Yes	Yes	Yes	Yes	Yes
Chemical and reagents available from multiple vendors?	No	Yes	Yes	No	Yes	Yes, *except NEB LAMP MM
Disclosure of primer sequences?	No	Yes std for PCR	Yes	No ProptryThermo	Yes	Yes
Primers commercially available from multiple vendors?	No	Yes	Yes CDC Primers	No ProptryThermo	Yes CDC SD Primers	Yes Available but not launched
Supply chain robust?	No	No/Maybe	Yes	No/Maybe	Yes	Yes
EUA Sponsor Organization Type	For Profit Company	Govt	Academic Not for Profit	Academic Not/For Profit ?	Public Benefit Corp	Public Benefit Corp
Designation of CLIA labs	Kit Sales	N/A open RoR	Impact & Expansion	Impact & Expansion	Impact & Expansion	Impact & Expansion

LAMP: Loop Mediated Isothermal Amplification

- Detects viral RNA
- LAMP generates 10X more DNA in 1/3 time compared to PCR
- Easy visual / camera-based readout



	qPCR	LAMP
Equipment	qPCR machine (thermal cycler & optics)	simple heat block or water bath (isothermal)
Time	90 min	30 min
Capital Cost	\$30K-100K	\$300



LAMP EUAS

EUA Date	Test	Diagnostic	Target	IC	Detection	Extraction	Amplification	LOD spike	LOD GE/ 3ml swab
Nov-17	Lucira (At Home POC)	Lucira COVID-19 All-In-One Test Kit	N, N	Lysis Ctrl + IC	Colorimetric	~Direct	Lucira device	Inactivated virus	2,700*
Oct-05	Seasun (2 duplex)	AQ-TOP COVID-19 Rapid Detection Kit PLUS	orf1ab, N	Human RNaseP	PNA Probe	Manual (Qiagen_60704 or Seasun_SS-1300) or Automated (Panagene_PNAK-1001 on PanaMax48)	BioRad_CFX96 or ABI7500	NCCP_43326 genomic RNA	3,000
Sep-01	Detectachem	MobileDetect Bio BCC19 (MD-Bio BCC19) Test Kit	N, E	only external controls	Colorimetric	Direct (1µl of transport media into LAMP)	heat block or qPCR (MD-Bio heater or BioRad_T100 or AB_Veriti or ...)	Twist_MT007544.1 synthetic gRNA	225,000
Aug-31	Mammoth	SARS-CoV-2 DETECTR Reagent Kit	N	Human RNaseP	CRISPR Probe	Automated (Qiagen_955134 on Qiagen_EZ1Advanced Benchtop)	ABI7500	SeraCare_AccuPlex_0505-0168 IVT encapsulated RNA	60,000
Aug-13	Pro-Lab	Pro-AmoRT SARS-CoV-2 Test	RdRP	only external controls	Probe	Direct (swab into 0.1ml) or kit (swab into 1ml then Pro-Lab_PLM-2000)	Optigene GenieHT	BEI_NR-52287 inactivated virus	125*
Jul-09	UCSF/Mammoth	SARS-CoV-2 RNA DETECTR Assay	N	Human RNaseP	CRISPR Probe	automated (Qiagen_955134 on Qiagen_EZ1)	ABI7500	SeraCare_AccuPlex_0505-0126 (lot 10480311) IVT RNA	60,000
May-21	Seasun (1 duplex)	AQ-TOP COVID-19 Rapid Detection Kit	orf1ab	Human RNaseP	PNA Probe	manual (Qiagen_60704)	BioRad_CFX96 or ABI7500	NCCP_43326 gRNA	21,000
May-18/ Aug31	Color	Color Genomics SARS-CoV-2 RT-LAMP Diagnostic Assay	N, E, nsp3	Human RNaseP	Colorimetric	automated (PerkinElmer_CMG-1033 on PerkinElmer_Chemagic360)	plate reader (Biotek_NE02), Hamilton Star	ATCC_VR-1986D (gRNA)	2,250
May-06	Sherlock BioSci	Sherlock CRISPR SARS-CoV-2 Kit	orf1ab, N	Human RNaseP	CRISPR Probe	Manual (ThermoFisher_12280050)	heat block or qPCR, Plate reader (BioTek_NE02)	gRNA	20,250
Apr-10	Atila BioSystems	IAMP COVID-19 Detection Kit	orf1ab, N	Human GAPDH	Probe	Direct (swab into 350µl, 15min RT then 3µl to LAMP)	BioRad_CFX96 or ABI7500 or Roche_LightCycler480II or Atila_PG9600	SeraCare_AccuPlex_0505-0129 pseudovirus (recombinant alphavirus)	3,500*

* [Lucira](#), [Pro-Lab](#) and [Atila](#) tests LOD/swab uses EUA specified volumes which are less than 3ml



LAMP!

RAPID RESULTS

New test for
koala chlamydia



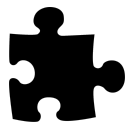
An Open Testing Future

Accessible, Scalable, Adaptable



Rapid test development

Test system quickly integrates new primers, and diverse sample media



Lab configuration

Modular design accommodates most any level of throughput and automation



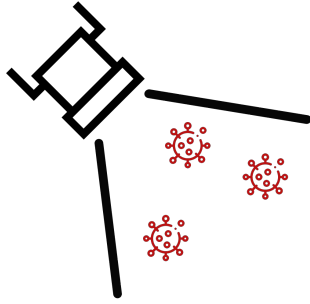
Amp chemistry

Isothermal & PCR proven to work. Emerging technologies can be added.



Performance flexibility

Mode to use case with pool, reflex, extraction, channels and seq as non core easy add ons



Thank You!

FloodLAMP Biotechnologies

A Public Benefit Corporation

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